

The Volatility Primer

Everything the field means by "volatility" — how it's modeled, how a model's forecasts get judged — walked through using your own BTC capstone as the running example, from the $R^2 \approx 0.99$ that turned out to be fake to the results that held up.

I What Volatility Actually Is

The theory every model in your notebook is trying to approximate.

Volatility is a measure of **how much** a price moves, not which way. Two assets can have identical average returns and wildly different volatility — one drifts up calmly, the other whipsaws to the same destination. Formally, it's usually expressed as the standard deviation of returns over some window: how dispersed the return distribution is around its mean.

REALIZED VOLATILITY

Computed *from observed price history* — e.g. the standard deviation of hourly returns over the trailing 7 days. Backward/concurrent-looking: a statistical estimate of how much the asset actually moved. This is what your entire capstone predicts (volatility_7d).

IMPLIED VOLATILITY

Backed out of an option's market price via a pricing model (Black-Scholes). Forward-looking: the market's own volatility expectation, priced in, and it embeds a risk premium. There is no options data anywhere in your project — this is why "IV" and "RV" got clarified early on.

WHY ANYONE FORECASTS IT AT ALL

Option pricing	Black-Scholes takes σ as a required input
Value-at-Risk	Loss estimates scale directly with σ
Position sizing	Trade smaller when choppy, larger when calm
Exchange margin	Collateral requirements rise with volatility

Predicting *direction* — where price goes next — runs straight into market efficiency: if it were easy, the edge would already be arbitrated away.

Predicting *volatility* is more tractable precisely because it doesn't ask which way price will move, only how rough the ride will be. That reframing is the whole premise of your project.

I.2 Six Stylized Facts

Empirical regularities every serious volatility model is built to capture — first catalogued rigorously in Cont (2001). Every model in Part II exists because of one or more of these.

01 Volatility clustering

Large moves are followed by large moves (of either sign); calm is followed by calm. Returns themselves show almost no autocorrelation, but *squared* returns (or absolute returns) do — a market that just had a violent day is more likely to have another violent day tomorrow, not because direction is predictable, but because *magnitude* is. This is the single fact GARCH's whole equation is built around.

02 Fat tails

Extreme moves happen far more often than a normal distribution predicts. This is why your GARCH cells fit with `dist='t'` (Student-t errors) rather than Gaussian — crypto returns in particular have heavy tails, and a fat-tailed error distribution fits the data's actual extremes instead of underestimating them.

03 Mean reversion

Volatility doesn't stay elevated (or depressed) forever — it reverts toward a long-run average over time. This is exactly the property that **breaks GARCH at your 168-hour horizon**: its forecast converges toward that long-run unconditional variance well before 7 days are up, which turns out to be a poor match for what `volatility_7d` actually does over that span.

04 The leverage effect

Volatility tends to rise *more* after a price drop than after an equally-sized rise. Named for a literal mechanical reason in equities (a falling stock price raises a firm's debt-to-equity ratio, i.e. its financial leverage), though the asymmetry shows up well beyond that original explanation. EGARCH and GJR-GARCH exist specifically to model this — plain GARCH(1,1) treats a +5% and -5% shock identically, which is structurally wrong.

05 Long memory

Autocorrelation in volatility decays *slowly* across many lags, rather than dropping to zero quickly the way it does for returns. Your own ACF plot confirmed this directly. HAR-RV exists because of this fact: instead of a complex fractionally-integrated model, it just averages realized volatility over three different horizons (day/week/month) and lets a simple linear model approximate long memory for free.

06 Cross-asset co-movement

Volatility regimes tend to move together across related assets, especially during systemic stress. Your ETH generalization test confirmed this is real for BTC/ETH — but your ETH-spillover-as-a-regressor check found the co-movement is *so* strong that ETH's realized vol adds no information beyond what BTC's own regressors already carry. Both findings are consistent with the same underlying fact.

II How Volatility Gets Modeled

Five model families, in the order your notebook builds them — each one exists to capture a specific stylized fact from Part I.

Baseline Naive persistence

Predict tomorrow = today: **the most recently known value, unchanged**. Not a strawman — it's the floor every real model has to clear. A model that can't beat "just copy the last known value" isn't adding anything.

Baseline Holt-Winters exponential smoothing

Forecasts a series as a weighted average of its own past, with a level, a trend, and a *seasonal* component. Included specifically to test whether hour-of-day/day-of-week seasonality (visible in EDA boxplots) was actually exploitable — its own seasonal smoothing parameter converged to $\gamma = 0.0000$, independently confirming what a chi-square test found separately: no.

Model 1 The GARCH family

Generalized Autoregressive Conditional Heteroskedasticity (Bollerslev, 1986, generalizing Engle's 1982 ARCH) — the econometric workhorse. Today's variance is a baseline level, plus a reaction to yesterday's shock, plus persistence from yesterday's variance:

GARCH(1,1)

$$\sigma^2_t = \omega + \alpha \cdot \epsilon^2_{t-1} + \beta \cdot \sigma^2_{t-1}$$

ω anchors the long-run level, α is the reaction to yesterday's squared shock, β is how much of yesterday's variance persists. Stationarity requires $\alpha + \beta < 1$ — the closer to 1, the more persistent (slower mean reversion).

EGARCH (NELSON, 1991)

Models $\log(\sigma^2)$ instead of σ^2 directly — no positivity constraints needed on parameters — and naturally builds in an asymmetric response to the sign of the shock, capturing the leverage effect.

GJR-GARCH (GLOSTEN-JAGANNATHAN-RUNKLE, 1993)

Adds one extra term that only activates on down-moves:
 $+\gamma \cdot \epsilon^2_{t-1} \cdot I(\epsilon_{t-1} < 0)$. A negative shock gets an explicit extra bump to variance. This is the spec your own AIC search selected as best.

Where it breaks in your project: GARCH's forecast decays toward its own long-run unconditional variance well before a 168-hour horizon is reached — a real, published structural property at long horizons, not an implementation bug. It posted $R^2 = -110.68$ on live data as a direct consequence.

Model 2 HAR-RV

Heterogeneous Autoregressive model of Realized Volatility (Corsi, 2009).

Rather than a complex long-memory model, it just regresses future volatility on realized volatility averaged over three different horizons — a cheap, linear way to approximate long memory:

HAR-RV

$$RV_{t+1} = \beta_0 + \beta_d \cdot RV_t^{(day)} + \beta_w \cdot RV_t^{(week)} + \beta_m \cdot RV_t^{(month)} + \epsilon$$

Estimated by plain OLS (or Ridge, in your case). Simple to fit, hard to beat empirically — which is exactly what happened in your own comparison.

Your HAR-RV-X extension adds one exogenous regressor: ATR_X , a scale-free (price-normalized) Average True Range term — an independent, range-based volatility proxy on top of the three RV-horizon terms. This is your project's winning model, and the fix that made it win (correcting atr_{14} 's raw-dollar scale to $atr_{pct} = atr_{14} / Close$) is covered in Part IV.

Model 3 EWMA / RiskMetrics

The simplest model here — a single decay parameter applied directly to squared returns, no lag columns, no feature engineering. This is the exact methodology J.P. Morgan's *RiskMetrics* (1996) popularized, and risk desks still reach for it:

EWMA

$$\sigma^2_t = \lambda \cdot \sigma^2_{t-1} + (1-\lambda) \cdot r^2_{t-1}$$

Today's variance forecast is a weighted average of yesterday's variance and yesterday's squared return. RiskMetrics' published default is $\lambda = 0.94$; your own backtested search over a wider grid selected $\lambda = 0.995$ for BTC.

Structurally, EWMA is an IGARCH(1,1) with no long-run anchor — **no mean reversion at all**. That's exactly why its one-step conditional estimate can stand in directly as its 168-hour-ahead forecast: there's no decay-toward-the-mean dynamic to get wrong at a long horizon the way GARCH does.

A deliberate exclusion Why not Random Forest, XGBoost, or an LSTM?

Every model above is **time-series-native by construction** — GARCH's equation is recursive and self-referential by definition, HAR-RV's regressors *are* volatility measured at different horizons, EWMA is a recursive filter. Generic tabular ML has no built-in notion of time at all: to XGBoost, each row is an unordered bag of numbers, and it only ever sees temporal structure through whatever lag columns an analyst hands it by hand.

Your project tested this claim directly rather than just asserting it — XGBoost, given the exact same lagged features plus explicit cyclical time features (`hour_sin/cos`, `dow_sin/cos`), still lost badly ($R^2 = -0.19$ vs. HAR-RV-X's 0.184). More flexibility without the right inductive bias just means more ways to overfit noise on a problem this hard.

III How a Model's Forecasts Get Judged

No single metric is trusted alone — each layer below protects against a different way a model can look good and not be.

RMSE

$$\sqrt{\text{mean}((\text{actual} - \text{pred})^2)}$$

Squares errors, so big misses count disproportionately more. Same units as the target. The most common metric for exactly this squared-penalty behavior.

MAE

$$\text{mean}(|\text{actual} - \text{pred}|)$$

Linear penalty — every error counts proportionally, no extra weight on big misses. More robust to a single catastrophic outlier than RMSE.

MAPE

$$\text{mean}(|\text{actual} - \text{pred}| / |\text{actual}|) \times 100$$

Scale-free percentage — easy to communicate, but blows up near zero. Your project's `evaluate()` guards this by excluding the smallest 1% of `|actual|`.

R^2

$$1 - SS_{\text{res}} / SS_{\text{total}}$$

"Variance explained vs. always guessing the mean." Unbounded below out-of-sample, and fragile: it depends on exactly which rows you test on, not just the predictions.

None of these are sufficient alone. A model can have a great RMSE and still be worthless (miscalibrated, no directional skill), or a misleadingly bad one (one outlier dominating). Your own project saw RMSE and MAE directly *disagree* about whether HAR-RV-X beat naive persistence on live data — that disagreement was itself informative, not a bug to resolve away.

The field's actual standard QLIKE

QLIKE (EVALUATED ON VARIANCE)

$$\text{QLIKE} = (\text{actual}/\text{pred}) - \ln(\text{actual}/\text{pred}) - 1$$

Multiplicative rather than additive — measures the *ratio* between predicted and actual, so a 2× miss counts the same whether it happens in a calm week or a chaotic one. RMSE, by contrast, gets dominated by whichever handful of high-volatility periods have the largest absolute errors.

Patton (2011) proved QLIKE is one of only two loss functions whose model rankings stay consistent even when the "actual" being scored against is itself a noisy proxy for true volatility — which realized volatility always is, since it's estimated from a finite sample of returns. RMSE has no such guarantee. QLIKE also penalizes *under*-predicting volatility more than over-predicting it, matching how a risk desk actually experiences the cost of being wrong.

Is the difference real, or noise? The Diebold-Mariano test

Two models can have different RMSEs purely by chance. The DM test asks directly: is Model A's edge over Model B statistically distinguishable from zero, given that their forecast errors are correlated with each other? For an h -step-ahead forecast, errors are serially correlated up to lag $h-1$ by construction — a naive t-test ignoring this will report false confidence. The correct version uses a Newey-West long-run variance with truncation lag $h-1$, plus the Harvey-Leybourne-Newbold small-sample correction.

At $h = 168$ hours, this correction is not a technicality — it's the difference between a headline result and an illusion. Your live-test sample of 5,496 predictions sounds large, but the effective sample size for a test this overlapping is closer to $5,496 / 168 \approx 33$ independent blocks. That's why HAR-RV-X's live edge (real in size) still isn't statistically proven yet.

Trustworthy scale, and direction Calibration & directional accuracy

MINCER-ZARNOWITZ REGRESSION

$$\text{actual} = \alpha + \beta \cdot \text{predicted} + e$$

(ideal: $\alpha=0$, $\beta=1$)

Checks whether a forecast's *scale* is trustworthy, not just its ranking. Your GARCH result had $\beta \approx 9.9$ historically — its forecast barely moves at all relative to reality, so you'd need to multiply it by $\sim 10\times$ just to line it up. That's the Mincer-Zarnowitz signature of GARCH's mean-reversion problem, visible in one number.

Directional accuracy — did the model correctly call whether volatility would rise or fall — has to be benchmarked against a proper baseline, not a 50/50 coin flip. Since volatility mean-reverts, a naive "it'll move toward its long-run average" caller already has a real edge over 50%. Your project's own mean-reversion baseline sits at 54–56%; a hit rate above roughly 62% against that baseline is treated as suspicious and investigated for leakage rather than celebrated.

The most dangerous failure mode Leakage, purging & embargo

A model can score brilliantly and still be measuring nothing real. The classic trap: a target window that overlaps almost entirely with information already available at prediction time. Your project's own opening discovery was exactly this — a 168-hour target sampled every hour shares 167/168 (99.4%) of its underlying returns with the value known one hour earlier, so "forecasting" it one hour ahead was close to copying, not predicting.

PURGED & EMBARGOED CROSS-VALIDATION

Even with a clean causal feature/target split, rows sitting right at a train/test boundary can still share overlapping information if the target itself spans a wide forward window. The fix (de Prado): drop a gap of rows — an *embargo* — around the boundary equal to the forecast horizon, so no train row's target quietly bleeds into what's nominally test-period data.

Your own audit found exactly this gap was missing (0.29% of train rows affected — small, but real, and now a documented recommendation).

The layer above statistics Economic viability

Everything above can be excellent and the model can still be worthless to trade. The final layer asks whether a real position built on the forecast makes money *after realistic costs*: Sharpe ratio, Sortino ratio, max drawdown, all computed net of transaction costs and slippage. There's also the **deflated Sharpe ratio** (Bailey & López de Prado) — the backtest-equivalent of a DM test, correcting a Sharpe ratio for how many strategy variants you tried before landing on this one.

This layer is **the one honest gap** in your capstone. There's no options data in the workspace to build a real trade against, so every result you have proves the *forecast* has statistical skill — not that trading on it would be profitable after costs. Those are different claims.

IV Your Capstone Project

The objective, in one sentence: predict Bitcoin's 7-day realized volatility, 168 hours ahead of the features used to predict it — using hourly BTC/USDT data from Binance via CCXT, evaluated against naive persistence and Holt-Winters, on both a historical split and genuinely later live data.

73,284

hourly candles, 2017–2026

168h

forecast horizon = window length

3

time-series-native models

5,664

live-test predictions logged

Regression, not classification: real downstream consumers of a volatility number — Black-Scholes σ , Value-at-Risk, position sizing — need the actual continuous value, not a high/low label.

Chronological The Investigation, In Order

This project's most important results weren't planned features — they were things that looked wrong, got investigated, and turned out to matter. In the order they actually happened:

THE RED FLAG

$R^2 \approx 0.99$ — too good to be real

At the original 1-hour-ahead design, even naive persistence scored R^2 near 0.99. Traced to a target-construction artifact: a 168-hour rolling target sampled every hour overlaps 99.4% with its own value one hour earlier.

THE FIX

Horizon redesigned to 168 hours

Setting the forecast horizon equal to the rolling window eliminates the overlap entirely — the standard non-overlapping design in the realized-volatility literature. Naive persistence and Holt-Winters immediately dropped to negative R^2 , exactly as they should once the target stops being trivially autocorrelated with itself.

THE CONSEQUENCE

Live-test window collapsed to ~48 hours

A 168-hour-ahead prediction can't be scored until 168 real hours elapse. Fixed by moving to fixed calendar boundaries — training through 2026-01-01, live test 2026-01-02 onward — expanding the live window to ~8 months.

A DESIGN SWAP

LSTM replaced with EWMA

The original third model slot had no econometric grounding for a volatility problem. Replaced with EWMA — an actual industry risk-desk tool with real theoretical footing, not a black box.

○ THE BIGGEST SINGLE FIX

ATR's ~45,000× scale bug

`atr_14` sat in raw dollar terms (mean $\approx$ \$303) against regressors on a ~ 0.001 – 0.03 fractional scale — worse, BTC's own price ranged \$4,000–\$90,000+ across the dataset, so a fixed dollar ATR meant something different at every point in history. Fixing it (`atr_pct = atr_14 / Close`) took HAR-RV-X's R^2 from -0.65 to $+0.18$.

○ INDEPENDENT VERIFICATION

The alpha audit

An independent script re-derived every prediction from raw data + saved models — not trusting the notebook's own printed numbers. Every live-test metric matched exactly. Found one new issue along the way: no purge/embargo gap at the train/test boundary.

○ GENERALIZATION TEST

The same model, frozen, tested on ETH

HAR-RV-X and EWMA's exact BTC-trained parameters — zero retuning — beat naive persistence on ETH too, with *stronger* significance than on BTC itself. Real evidence the signal is structural to crypto volatility, not a BTC-history fluke.

○ THE R^2 CEILING

5-minute resolution, tested properly

A volatility-signature-plot sanity check ruled out microstructure noise down to 5 minutes for BTC/USDT on Binance. A scoped HAR-RV-X migration then tripled live R^2 ($0.061 \rightarrow 0.206$) — though it didn't help on the historical split, an honestly mixed result, not a clean win.

Final numbers Models Built & Results

MODEL	HIST. RMSE	HIST. R ²	LIVE RMSE	LIVE R ²
HAR-RV-X — Tuned	0.001418	0.1840	0.001502	0.0610
EWMA ($\lambda=0.995$) — Tuned	0.001532	0.0593	0.001540	0.0133
Naive Persistence	0.001649	-0.0897	0.001676	-0.1686
Holt-Winters	0.002845	-2.2448	—	—
GJR-GARCH(t) — Tuned	0.003876	-5.0225	0.016385	-110.68

HAR-RV-X wins on every metric, on both splits. GARCH is the clear laggard — not from an implementation bug, but the structural mean-reversion property covered in Part II.

The honest ledger What Moved the Needle — and What Didn't

Six systematic improvement attempts, reported whichever way they went — the negative results are as methodologically important as the positive ones.

CHANGE TESTED	RESULT
ATR scale fix (atr_pct)	Adopted R^2 $-0.65 \rightarrow +0.18$
Wider EWMA λ search grid	Adopted λ $0.94 \rightarrow 0.995$
RSI / EMA spread / BB width / MACD as extra regressors	No help R^2 moved <0.002
Log-transforming the target	No-op values too small for log1p to matter
Fixed HAR-RV-X + EWMA ensemble	Failed CV selected weight $\rightarrow$ 100% HAR-RV-X
XGBoost + explicit time features	Lost badly $R^2 = -0.19$ vs. HAR's 0.184
ETH volatility as a spillover regressor	Actively hurt R^2 $0.184 \rightarrow 0.112$

The most important finding here might be the ensemble failure. A naive look at test-set performance suggested a HAR+EWMA blend was a genuine improvement — but the CV-selected weight converged to 100% HAR-RV-X once tuned honestly on training data only. Same category of mistake as the original $R^2 \approx 0.99$ artifact, caught the second time only because rolling-origin CV was applied rigorously instead of skipped as "obviously fine."

Where things actually stand **The Verdict**

HAR-RV-X shows a real, statistically significant edge over naive persistence on the historical test (DM $p=0.0125$). On the live/out-of-sample test — the one that matters most — the edge is the same *size* but not yet statistically *provable* (DM $p=0.24$): 5,496 live observations isn't enough given how much a 168-hour-ahead test overlaps with itself. That's not a failure — it's an honest, calibrated "promising, not yet proven," which is a more useful capstone result than an artificially perfect one would have been.

0.1840

HAR-RV-X historical R^2

0.0610

HAR-RV-X live R^2

$p=0.0125$

significant historically

$p=0.24$

not yet significant, live

Still open: the economic/PnL layer (no options data to test against), the full 5-minute pipeline migration (scoped result only), and genuinely new information sources — on-chain metrics, news sentiment, Realized GARCH / HAR-RV-Q — none of which this workspace has data for yet.

Compiled from the BTC volatility capstone's notebook, report, alpha audit, and every follow-on experiment run alongside it. Every number above was re-checked against a script output before being written here.